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RESEARCH ARTICLE

Next-Generation Text Summarization: A T5-LSTM FusionNet Hybrid Approach for Psychological Data

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ABSTRACT Automatic text summarization (ATS) has developed as a vital method for compressing massive amounts of textual content into concise and useful summaries, to retrieve more effective and useful information. ATS reduces textual statistics into coherent and shorter versions especially focusing on psychological text summarization to extract insights and emotional states, assisting in better analysis and understanding of psychological contents. In this context, this study proposes a new hybrid model T5-LSTM FusionNet, to enhance textual content summarization in the field of psychology. The motivation derives from the developing extent and accessibility of psychological literature online, which necessitates exceptional techniques for extracting significant findings quickly and reliably. The recommended T5-LSTM FusionNet model combines the benefits of Text-to-Text Transfer Transformer (T5) and Long Short-Term Memory (LSTM). The dataset with 5480 records, accumulated from numerous psychology-associated websites, has been used to verify its performance. T5-LSTM FusionNet's overall performance is evaluated against several latest models along with T5, LSTM, BERT, and DistilBERT. Measures such as accuracy, precision, recall, F1-score, and ROUGE rankings are used to evaluate the model's exceptional summarization. With T5-LSTM FusionNet accomplishing a precision of 0.72, recall of 0.72, F1-score of 0.71, and accuracy of 0.74, the effects show significant improvement over individual models like T5 and LSTM, as well as competitive models like BERT and DistilBERT, in terms of summarization effectiveness and accuracy. Furthermore, T5-LSTM FusionNet plays thoroughly in catching both unigram and bigram overlaps concerning summaries, as proven through a comparison study using ROUGE metrics. This suggests that T5-LSTM FusionNet can retain sequence integrity and relevance in summarizing tasks. This work advances ATS techniques in psychology by presenting a hybrid model that combines sequential and transformer-based learning strategies.

INDEX TERMS Automatic text summarization, T5-LSTM FusionNet, psychology, natural language processing, machine learning.

I. INTRODUCTION

The text summarization process is important when shortening a piece of the text while keeping the original text's important meaning and informational features as it is. As in the past, manual text summarization was time-consuming and difficult, however, the automation of text summarization is becoming an important and prevalent part of academic research and studies [1]. Automatic text summarization (ATS) reduces the lengthy text without losing the main idea and relevant information from the original text [2]. ATS aims to reduce the large volume of textual information to an

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articulate version, shorter to make it easy for the reader to get the main idea quickly. Previous studies focus on different text summarization techniques in various domains, however, this study especially focuses on psychological text summarization to highlight and extract the main and important psychological insights and emotional states from the text. Aiding in better analysis and understanding of the psychological contents.

Through natural language processing (NLP), this process is difficult but worthwhile. It pursues to compress information from huge texts into shorter forms, making retrieval and analyzing less complicated even as retaining important nuances. This permits users to apprehend the main thoughts while maintaining important data swiftly. The accessibility of information, such as articles, news, and e-book critiques, has expanded affectedly over the past two decades due to the fast development of the Internet [3], and this tendency continues to grow at a short pace. The volume of literature available online is rising rapidly, and users frequently seek queries to locate unique content online. Users nonetheless must visit several websites, that's annoying and time-consuming. Text summarizing is a practice that has been proposed to address this issue. It challenges the exertion of organizing large volumes of material and quickly removing important details from protracted publications [4].

ATS is divided into two main categories, a) Extractive summarization which aims to separate relevant passages from the original text, commonly using linguistic or statistical methods [5], and b) Abstractive summarization which goes outside just repeating passages from the original text, instead, it rewords the text to produce new phrases that might not be found in the original text [6]. Abstractive summarization is more difficult than extractive summarization when generated through traditional machine learning (ML) and NLP methods since it is important to interpret and do a semantic analysis of the text to provide an abstractive summary [7].

The importance of text summarization can be found in its applications in various domains including information retrieval, Google News, document summarization, content production, Text Teaser, Summaries Bot, news aggregation, IBM Watson, Microsoft, language translation, and clinical text summarization [2]. Clinicians, in particular, devote significant effort to summarizing vast textual material, such as assembling diagnostic reports, writing progress notes, or synthesizing a patient's treatment history across many experts [8], [9]. This endeavor is difficult even for seasoned clinicians with extensive experience [10], [11]. However, this study focuses on automatic text summarization (ATS) in psychology as a customer service support system (C3S). To the best of our knowledge, this is the first study that has been done in this field. The first task in the C3S is to summarize text relating to the domain. The user may give large paragraphs or text documents that must be summarized to be easily classified based on the required task, which in this case is disorder classification. The study focuses on psychological data collected from several online sources. A new ensemble model, T5-LSTM FusionNet, is presented for summarization and is based on Text-to-Text Transfer Transformer (T5) and Long Short-Term Memory (LSTM). The proposed model is compared with other state of the art model to strengthen the study based on standard assessment criteria including, precision, recall, f-measure, accuracy and Recall-Oriented Understudy for Gisting Evaluation (ROUGE). The benchmarked models are T5, LSTM, Bidirectional Encoder Representations from Transformers (BERT), and DistilBERT. This study makes several significant contributions to the field of ATS, especially in the psychological arena which are:

- Presents T5-LSTM FusionNet, a unique hybrid model that combines the language modeling capacity of T5 with the capability of LSTM to capture sequential interactions.
- Especially design models to improve the accuracy and effectiveness of text summarization tasks in the psychological domain.
- Evaluated against well-known models, including LSTM, BERT, DistilBERT, and T5.
- Makes use of a dataset of 5480 records from several websites concerning psychology, ensuring data quality through extensive preparation.
- Assesses a unique methodology designed to enhance cutting-edge ATS techniques for improved understanding and classification of psychological data.

The proposed work aims to improve the accuracy of summarizing psychological data by using a hybrid T5-LSTM FusionNet model. This model combines the generative abilities of T5 (a Transformer-based model) with the sequential learning strengths of LSTM, which is particularly well-suited for capturing the context within psychological narratives. The summarization result effectively condenses complex psychological texts into shorter, coherent versions while preserving essential insights, enabling mental health professionals to analyze data more efficiently and make timely interventions based on key psychological cues.

Psychological data was selected for this study because it contains rich, narrative-driven insights into individuals' mental states, often requiring nuanced interpretation. Summarizing this data effectively can assist mental health professionals in capturing significant information quickly, thus improving both the accessibility and usability of complex reports. Through this summarization, our model can detect and highlight critical patterns or shifts in mental health that are essential for effective treatment planning, making the summarized data valuable in identifying trends or anomalies in patient conditions over time.

This study's remaining sections are organized as follows: A review of relevant literature is provided in Section II, including earlier studies in the area. The research strategy and technique used in this study are described in Section III. An analysis of the findings and a discussion of their ramifications are presented in Section IV. The conclusions derived from the study's findings are finally presented in Section V.

II. RELATED WORK

The two main categories of automatic text summarizing techniques are extractive and abstractive approaches. To communicate the main ideas of the original text, abstractive summarization generates new terms [12]. This approach seeks to recognize the means of the text, requiring a large amount of manually labeled training data. Abstractive strategies can be hard to implement, and the summaries they provide are sometimes tough to examine and recognize [13]. Conversely, extractive summarization minimizes duplication and prevents grammatical problems by selecting and extracting phrases directly from the textual content. Methods based on statistical techniques, feature scoring, and subject models are examples of extractive summarization techniques.

NLP has appreciably investigated strategies for textual content summarization. Researchers have advanced several strategies for condensing enormous amounts of textual content into summaries, every with its technique. Traditional techniques include frequency-based algorithms developed by [14], [15], [16], which prioritize words based totally on their frequency and place. Metaheuristic approaches, first defined through s and later modified by way of Saini et al. [17] are searching to optimize the summarization of the usage of algorithms that imitate natural strategies. Bayesian techniques, as described by Vanderwende et al. [18] and Ma and Nakagawa [19], use probabilistic models to enhance the accuracy and relevance of summaries.

Graph-based techniques, such as Textrank [20] and Lexrank [21], consider phrases as nodes with edges expressing their similarity ratings. These algorithms excel at identifying critical sentences for a summary. Clustering approaches have also proved useful in summarizing attempts. Researchers have investigated clustering-based techniques [22], [23], which group similar phrases together to discover essential material. Some research has just focused on clustering algorithms, such as K-means, which identify relevant phrases for summarization [24], [25]. Others have combined clustering and optimization approaches [26], which improve summary quality by assuring complete coverage and eliminating redundancy.

Until recently, studies focusing on other legal document types like bills, reports, and patents [27], [28], [29] have mostly compared documents from case judgments [30], [31]. Growing interest has been shown in integrating deep learning models into legal document summarizing systems since their introduction [32]. Thanks to developments in NLP and computational methodologies, text summarization is becoming more and more versatile to meet a wide range of demands in different document types and areas. Compared to single-document summarizing, multi-document summarization presents more significant obstacles [33]. It needs to manage several papers with redundant data, condense large amounts of text, and guarantee the effective extraction and selection of phrases [2], [34]. Statistical methods and optimization techniques are used to address these issues [35]. One of the most important challenges facing any ATS system is controlling duplication and relevancy in the summarized content [36].

Text summaries are produced using subject model-based techniques, which identify hidden themes in the text and examine subjects at the sentence level. These suggested themes have been identified in large part because of the Latent Dirichlet Allocation (LDA) topic model, which is based on probabilistic latent semantic analysis (PLSA) [37]. However, in dealing with several parameters, PLSA frequently has overfitting issues. LDA uses the Dirichlet distribution as the prior for the document-topic and word-topic distributions and adds hyper parameters to address this. An adaptation of LDA that establishes hierarchical topic links using a nonparametric method based on the layered Chinese restaurant process was developed by Blei et al. [38], [39]. Another study [40] successfully used this technique in English texts, with encouraging outcomes. Furthermore, Arora et al. improved sentence computations by combining LDA with singular value decomposition, which improved the summarization's capacity to communicate subject meaning [41]. By using the LDA model in conjunction with the creation of a subject hierarchy, Akhtar created a technique for summarizing news [42]. Sriramya [43] introduce the Ontology-based Knowledge Aware Multi-focus Conditional Generative Adversarial Network (OKAM-CGAN) for Abstractive Text Summarization. Their model outperforms existing methods, achieving ROUGE-L scores of 42.1% and 45% on the CNN/Daily Mail and Edmunds datasets, respectively, demonstrating its effectiveness in summarizing large documents. Garg et al. [44] introduce OntoDSumm, an ontology-based method for summarizing disaster-related tweets. By leveraging domain knowledge in both categorization and ranking, OntoDSumm significantly outperforms six state-of-the-art methods, achieving a 2%–66% improvement in ROUGE-1 F1-score across 12 disaster datasets, thereby enhancing real-time disaster response efficiency.

Ayed et al. [45] present a two-stage cognitive abstractive summarization model. The first stage maps text segments to salient topics and computes saliency scores, while the second uses fuzzy logic to select sentences based on coverage and fidelity. The model then rephrases these sentences with a T5 transformer, demonstrating superior performance compared to three state-of-the-art summarization protocols. Table 1 presents the summary of related work.

The emphasis and technique of the proposed study set it apart from previous research. In contrast to other studies that focused on different document categories such as invoices, reports, patents, and legal papers, this study focuses on the field of psychology, which has not been well studied in prior text summary research. Furthermore, this study presents a novel hybrid model called T5-LSTM FusionNet, whereas previous methodologies primarily used either extractive or abstractive summarization techniques, with many concentrating on conventional methods like frequency-based

algorithms, metaheuristic approaches, Bayesian techniques, graph-based methods, and clustering approaches. Through the use of an ensemble technique, this model combines the sequential learning features of LSTMs with the sophisticated language interpretation and generating skills of T5, to improve summarization accuracy. This novel method is tested against the most recent models, including T5, LSTM, BERT, and DistilBERT. Its efficacy is also thoroughly compared and validated using standard evaluation metrics, such as precision, recall, F1 score, accuracy, and ROUGE.

As compared with the previous studies mentioned in the literature, the T5-LSTM FusionNet addresses the key challenges related to existing hybrid models, in terms of computational overhead and semantic gaps. Traditional hybrid models face high resource consumption arising from the complexity of mixing up different architectures. It improves performance by using the capabilities of both Transformer-based (T5) and sequential models, in terms of balancing up computational demands. It renders a strong contextual understanding through T5, while LSTM captures sequential dependencies, thus reducing the semantic gap that often occurs in models relying solely on one architecture. Therefore, this mixed approach will certainly ensure both contextual richness and sequential integrity, offering an even better compromise solution for text summarization and analysis.

III. RESEARCH DESIGN AND PROCEDURE

This study aims to propose a new solution for ATS to produce better accuracy as compared with previously proposed models in the recent past. Moreover, this study focuses on the text summarization problem in the domain of psychology. A new hybrid model T5-LSTM FusionNet has been proposed in this study for text summarization. The proposed model is benchmarked with state-of-the-art models including T5, LSTM, BERT, and DistilBERT. The models are evaluated using standard assessment measures which are precision, recall, F1 score, accuracy, and ROUGE. Various steps are involved in this process including data collection, cleaning, model training, and evaluation. The overall process is shown in Figure 1, described in the subsequent accordingly.

We used a "random search" strategy to find the best values for each model separately, focusing on three important hyperparameters: training batch size, learning rate, and number of epochs. To determine which configuration produces the best results on the validation dataset, this approach entails randomly sampling several combinations of these hyperparameters within preset ranges [46]. We selected the training batch size to be between 16 and 128 and the range of epochs to be between 5 and 30, as well as the learning rate to be between 0.00001 and 0.0001. For every model, we performed ten or random search iterations. Accuracy metrics were used to assess each setup during this procedure. For every model, the best configuration was determined by taking the one with the highest accuracy. Table 2 provides a summary of our hyperparameter optimization outcomes and lists the chosen hyperparameters for each fine-tuning model.

A. DATA SELECTION AND PREPROCESSING

The data used in this study has been collected from various sites[1,2,3,4,5,and6] related to the domain of psychology. The final dataset comprises 5,480 records, primarily focused on data related to cognitive behavior, emotional responses, and mental health assessments. These categories were selected to align with the study's objectives of analyzing emotional sentiment, as well as Anxiety, Depression, Loneliness, and Stress. The collected data underwent multiple preprocessing steps to ensure suitability for training and testing the applied models.

In the data preparation phase, various essential steps take place to prepare the text data for successful model training and assessment as shown in Figure 2.

First, contractions are enlarged, transforming phrases such as "isn't" to "is not," to guarantee text clarity and consistency. To ensure uniformity and eliminate word conflicts caused by case variations, all content is converted to lowercase. However, to avoid contradictions when dealing with named entities and acronyms, specialized techniques are applied. Named Entity Recognition (NER) is used to detect and preserve the original case of named entities and acronyms, ensuring that important terms remain accurate. Additionally, context-aware tokenization is implemented to handle special cases, such as acronyms like "US," allowing them to retain their proper capitalization while other parts of the text are normalized. These combined approaches ensure that key terms are preserved without compromising the overall consistency of the content. HTML tags are deleted to clean up any web-related markup, leaving just useful text content. Text in brackets and brackets is removed to concentrate on the primary point and exclude extraneous information that may not add to summarizing. Possessive ends are deleted by removing the "'s," simplifying the text structure without changing the meaning. Non-alphabetic letters are deleted, which helps to reduce noise while maintaining just significant textual information. English stopwords, such as "and," "the," and "is," are filtered away to highlight the text's most important terms. In the last, the null value from the data is removed. These preprocessing methods provide a consistent, clean, and focused dataset which improves the performance of text summarization models. After initial filtering, the data were divided into

[1] https://aclanthology.org/I17-1099/
[2] https://github.com/facebookresearch/EmpatheticDialogues
[3] https://github.com/rkadlec/ubuntu-ranking-dataset-creator
[4] https://github.com/facebookresearch/ParlAI/tree/main/parlai/tasks/personachat
[5] https://www.cs.cornell.edu/~cristian/Cornell_Movie-Dialogs_Corpus.html
[6] https://www.isaacmshamam.com/blog/emotional-reasoning-a-dialogue-between-a-therapist-and-patient

TABLE 1. Comparison of automatic text summarization technique: methods, advantages, and challenges from the related work.

Technique Category	References	Description	Models	Advantages	Challenges
Extractive Summarization	[27], [28], [29]	Selects and extracts sentences or phrases directly from the text based on statistical techniques.	Textrank, Lexrank	Retains original wording, avoids grammatical errors	May miss out on generating new information
Abstractive Summarization	[12], [13]	Generates new phrases to convey the main ideas, often using deep learning and natural language generation techniques.	LDA-based methods, deep learning models	Can generate novel summaries, potentially more readable	Requires large amounts of training data, harder to implement reliably
Frequency-Based Algorithms	[14], [15], [16]	Prioritizes words based on their frequency and position in the text.	-	Simple to implement, computationally efficient	Limited in capturing context and semantic meaning
Metaheuristic Approaches	[17]	Uses algorithms inspired by natural processes to optimize summarization.	-	Can find optimal solutions, adaptable to various domains	May require fine-tuning, less interpretable
Graph-Based Techniques	[20], [21]	Treats words as nodes in a graph, using edges to express similarity ratings.	Textrank, Lexrank	Effective at identifying important sentences	Sensitivity to graph construction parameters
Clustering-Based Techniques	[22], [23], [24], [25], [26]	Groups similar phrases together to identify key content.	K-means clustering, other clustering algorithms	Handles redundancy well, identifies thematic clusters	Dependency on clustering quality and algorithm choice
Deep Learning Models	[32]	Integrates neural networks to learn representations for summarization.	Seq2Seq models, Transformer-based architectures	Capable of learning complex patterns, context-aware summarization	Requires large datasets, computationally intensive
Bayesian Techniques	[18], [19]	Uses probabilistic models to enhance summary accuracy and relevance.	Variational methods, Bayesian inference	Provides uncertainty estimates, robust to noise	Complex model tuning, computational cost
Topic Modeling Based	[37], [38], [39], [40], [41], [42]	Identifies hidden themes and analyzes subjects at the sentence level.	LDA, PLSA, Hierarchical topic models	Effectively captures themes and subject meaning	Overfitting issues, parameter tuning challenges, computational complexity
Multi-Document Summarization	[2], [33], [34], [35]	Summarizes multiple documents by managing redundancy and ensuring comprehensive coverage.	Optimization-based methods	Handles large volumes of data, synthesizes information from multiple sources	Increased complexity, integration of diverse document types
Ontology-Based	[43], [44]	OKAM-CGAN, OntoDSumm using ontology-based methods.	OKAM-CGAN; OntoDSumm	Generates novel summaries and enhances the information retrieval.	OKAM-CGAN may require extensive training data.

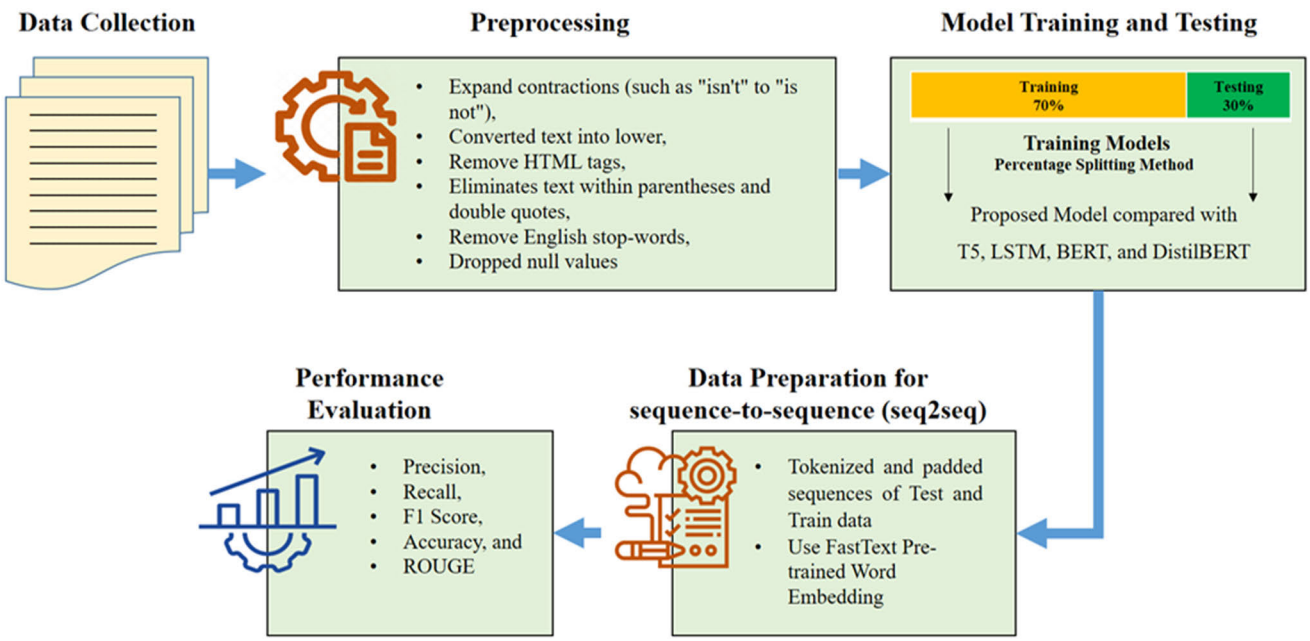

FIGURE 1. Workflow of the research study for text summarization in psychology.

TABLE 2. Summary of hyperparameter optimization outcomes and chosen hyperparameter for each fine-tuning model.

Model	Training Batch Size	Learning Rate	Number of Epochs
T5	32	0.00005	20
LSTM	64	0.00007	15
BERT	48	0.00003	25
DistilBERT	16	0.00010	10
T5-LSTM FusionNet	128	0.00008	30

training and testing sets to ensure the model's robustness and generalization. An 70/30 split was used for this division.

B. DATA PREPARATION FOR SEQUENCE-TO-SEQUENCE (SEQ2SEQ) TASK

A systematic strategy consisting of tokenization, padding, and FastText pre-trained word embeddings is used to prepare the data for sequence-to-sequence tasks in text summarization. In the first phase, the training and testing data are tokenized into a series of numbers, with each integer representing a word or token from the vocabulary. Tokenization converts each word x_i to an integer t_i, resulting in:

$$T = \{t_1, t_2, \ldots t_n \tag{1}$$

In the second phase, the tokenized sequences are padded to guarantee that they are of the same length. It includes adding a special token (usually zero) to sequences that are less than the maximum length, which results in sequences of the same

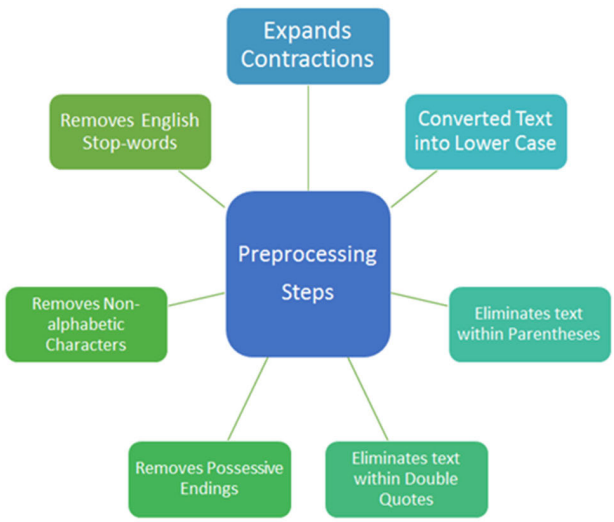

FIGURE 2. Preprocessing steps used in this study for text summarization task.

length l. For instance, a tokenized sequence $\{t1, t2, t3\}$ with a maximum length $l=5$ is padded to $\{t1, t2, t3, 0, 0\}$.

In the next phase after tokenization and padding, the FastText pre-trained word embeddings are used to transform tokens into a dense vector representation. FastText embeddings identify semantic links between words by encoding each word t_i as a dense vector $v_i \in R^d$, where d is the embedding dimension. These embeddings are discovered from massive corpora and include syntactic and semantic commonalities, which enhance the model's comprehension of the textual content.

The embeddings are then fed into the seq2seq model. The embedded sequence $\boldsymbol{E}$ for a tokenized and padded sequence $\boldsymbol{T}$ is written as:

$$E = \{v_1, v_2, \ldots v_n\}, \tag{2}$$

where $\boldsymbol{v_i}$ is the embedding vector for token $\boldsymbol{t_i}$. The seq2seq model's encoder receives this embedding sequence E and analyses it to produce a context vector. The decoder then utilizes this context vector to construct the output sequence, which is a condensed version of the original text. This rigorous data preparation procedure guarantees that the model receives well-structured, semantically rich input, which improves its capacity to provide accurate and coherent summaries.

C. MODEL BUILDING (T5-LSTM FUSIONNET)

T5-LSTM FusionNet's architecture for ATS uses the advantages of T5 and LSTM fashions in the use of an ensemble technique. Firstly, each of the T5 and LSTM fashions is initialized. For each record, the tokenized and padded collection is first processed through the T5 encoder to generate a context vector C_{T5}. Simultaneously, the embedded sequence is processed by using the LSTM model, which produces a hidden state vector H_{LSTM}. These two vectors, C_{T5} and H_{LSTM}, are then fused right into a single context vector, C_{Fusion}, using a fusion feature that can encompass concatenation, averaging, or a neural network-based approach. The fused context vector C_{Fusion} is then sent into the T5 decoder, which produces the summarized textual content *Si* for every page. This technique is performed for each document in the dataset, resulting in a set of summaries. The combination of T5's excellent language knowledge and generating capabilities with LSTM's potential to capture sequential relationships yields a very good model for textual content summarization. Algorithm 1 provides an outline of the T5-LSTM FusionNet architecture.

The T5-LSTM FusionNet architecture for textual content summarization combines several sophisticated components and settings to enhance performance. To capture and encode sequential dependencies inside the textual content, the LSTM thing incorporates a 3-layer LSTM encoder with 513 dimensions per layer. The decoder consists of a single LSTM layer with an interesting mechanism that lets the model focus on critical sections of the input series whilst generating summaries. This attention mechanism improves the decoder's capacity to provide coherent and context-relevant summaries.

Text input that has been tokenized and padded is to start with processed through the T5 encoder to provide the context vectorC_{T5}. Concurrently, the same sequences are embedded with FastText pre-trained embeddings of 300 dimensions, changing every token right into a dense vector representation. These embedded sequences are then processed with the aid of the 3-layer LSTM encoder to generate the hidden nation vector H_{LSTM}. C_{T5} and H_{LSTM} are fused right into a single context vector C_{Fusion} the usage of a specific fusion function,

Algorithm 1 T5-LSTM FusionNet for Text Summarization

Input:

- Tokenized and padded sequences of text data $D = \{d_1, d_2, \ldots d_n\}$
- FastText pre-trained word embeddings

Output:

- Summarized text for each document in $\boldsymbol{D}$

Steps:

1. Model Initialization:

1.1. Initialize T5 model M_{T5}.

1.2. Initialize LSTM model M_{LSTM}.

2. Encoding Phase:

2.1. For each document $\boldsymbol{d_i}$:

2.1.1. Pass the tokenized and padded sequence through M_{T5} encoder to get context vector C_{T5}.

2.1.2. Pass the embedded sequence through M_{LSTM} to get a hidden state vector H_{LSTM}.

3. Fusion Phase:

3.1. Combine C_{T5} and H_{LSTM} into a single context vector C_{Fusion}.

$$C_{Fusion} = (C_{T5}, H_{LSTM}) \tag{3}$$

where Fusion is a function that combines both vectors (e.g., concatenation, averaging, or a neural network-based method).

4. Decoding Phase:

4.1. Use the combined context vector C_{Fusion} as input to the T5 decoder to generate the summarized text Si for document $\boldsymbol{d_i}$.

5. Output:

5.1. Return the summarized text $\boldsymbol{S}$ for each document in $\boldsymbol{D}$.

inclusive of concatenation, averaging, or a neural network-based method.

The merged context vector C_{Fusion} is then passed into the T5 decoder, which generates the summary text. The model is optimized with the *RMS* optimizer, which is properly appropriate for coping with the non-stationary desires that are common in neural network training. The loss function employed is sparse categorical cross-entropy, which is suitable for multi-class classification tasks such as text creation. This complete architecture uses T5's robust language modeling skills, as well as LSTMs' sequential learning strengths, which are reinforced by an attention mechanism, to provide better text summarization.

Concatenation would be used where the full representational capacity of both C_{T5} and H_{LSTM} is to be preserved, as this would make the model capitalize on the wide variety of features from both the embeddings. In cases where one needs to reduce the dimension and ensure that each source contributes balanced, thereby preventing overfitting and making generalization more effective, averaging is used. The neural network-based approach, with a learnable transformation

layer typically used, is applied in scenarios where the relation between the two embeddings is more complex and requires adaptive weighting for optimal performance. The choice of these approaches depends on empirical comparisons and the specific needs of the downstream task.

Fine-tuning was carried out on the T5 as well as the LSTM models, according to our task-specific dataset to improve performance. In the fine-tuning process, task-specific prompts were used to train the T5 model, along with a low learning rate to prevent catastrophic forgetting during adjustment. Attention was modified to only fine-tune the last few transformer layers and freeze earlier layers to ensure general language understanding. The fine-tuning process for the LSTM model consisted of adding pre-trained word embeddings, changing the number of hidden units, and adding dropouts to avoid overfitting. Training of both models also used a loss function which we had adjusted for our objectives. Other parameters like batch size and optimizer parameters were also chosen such that we achieve better results.

D. BENCHMARKED MODELS AND PERFORMANCE EVALUATION

The proposed model is compared with the state-of-the-art model listed in Table 3 based on standard assessment criteria including precision, recall, F1 score, accuracy, and ROUGE [2]. These measures can be calculated as:

$$Precision = \frac{TP}{TP + FP} \tag{4}$$

$$Recall = \frac{TP}{TP + FN} \tag{5}$$

$$F1 - Score = \frac{2 * Precision * Recall}{Precision + Recall} \tag{6}$$

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{7}$$

where the rate of true negative values (TN), true positive values (TP), false positive values (FP), and false negative values (FN) are expressed by the equations.

- True Positive (TP): A summary generated is classified as a TP if it includes relevant and accurate information that are present in the reference summary.
- True Negative (TN): As summarization is a generative task, it is quite challenging to define TN. In our evaluation, TN refers to cases where the model correctly excludes information that is not in the reference summary as incorrect or hallucinated.
- False Positive (FP): The generated summary has incorrect, redundant, or even hallucinated information not present in the reference summary.
- False Negative (FN): FN refers to those cases where key information from the reference summary is missing in the generated summary.

$$ROUGE - N = \sum gram \in RefSmin \tag{8}$$

TABLE 3. List of benchmark models used in the study.

Benchmarked Models	References
T5	[47], [48], [49]
LSTM	[50], [51], [52]
DistilBERT	[53], [54], [55]
BERT	[33], [56], [57]

where $RefS$ is the reference summary. $Count_{match}(gram)$ is the number of n-grams in the candidate summary that match an n-gram in the reference summary. $Count_{cand}(gram)$ is the total number of n-grams in the candidate summary. $Count_{RefS}(gram)$ is the total number of n-grams in the reference summary.

$$ROUGE - L = F_{LCS} = \frac{\left(1 + \beta^2\right).R_{LCS}.L_{LCS}}{R_{LCS} + \beta^2.P_{LCS}} \tag{9}$$

where, $R_{LCS} = \frac{LCS(X,Y)}{Length(Y)}$ is the recall of the LCS, $P_{LCS} = \frac{LCS(X,Y)}{Length(X)}$ is the precision of the LCS,

β is typically set to 1.

IV. RESULTS ANALYSIS AND DISCUSSION

This section discusses the experimental outcomes achieved via each employed model. As the study aims to propose a new model for ATS, a hybrid model T5-LSTM FusionNet has been introduced and compared with state-of-the-art models listed in Table 3. These models are compared based on precision, recall, F1-score, accuracy, and ROUGE. Figures 3-5 present the analysis achieved through ROUGE1, ROUGE2, and ROUGE L, where, ROUGE-1 focused on individual words, ROUGE-2 on pairs of consecutive phrases, and ROUGE-L at the longest collection of matching phrases, providing a comprehensive evaluation of the generated summaries from unique perspectives.

The operations applied to the psychological dataset, particularly through the T5-LSTM FusionNet model, are necessary to handle the data's complex and sequential nature. The model operations, such as encoding, attention mechanisms, and LSTM's sequential processing, enable the transformation of lengthy psychological text into coherent summaries without losing critical details. Table 4 shows the sample results demonstrate the model's proficiency in generating concise summaries that retain the intended psychological interpretations, ultimately aiding practitioners in making faster and more informed decisions.

Figure 3 shows the outcomes assessed through ROUGE 1 where the proposed model shows better performance as compared with the rest of the employed models with the value of 0.662. The T5 model has a ROUGE-1 score of 0.21, indicating its relatively lower effectiveness in this task. Even as LSTM performed a slightly better score of 0.242. DistilBert is significantly better with a score of 0.35. BART, known for its powerful text technology capabilities, scored 0.456, demonstrating its proficiency in generating

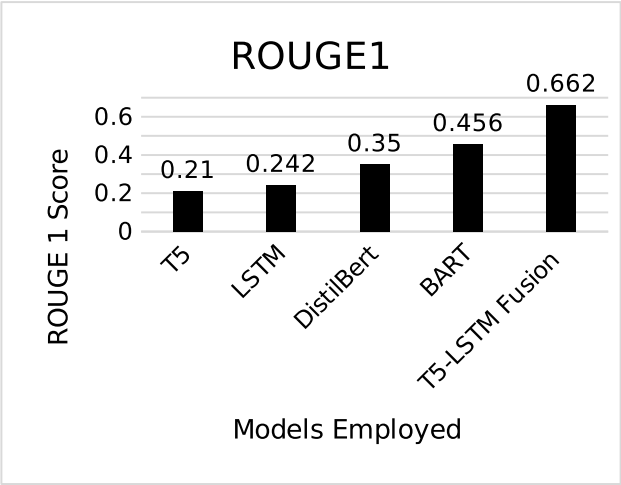

FIGURE 3. ROUGE-1 Score comparison across the proposed and other employed models.

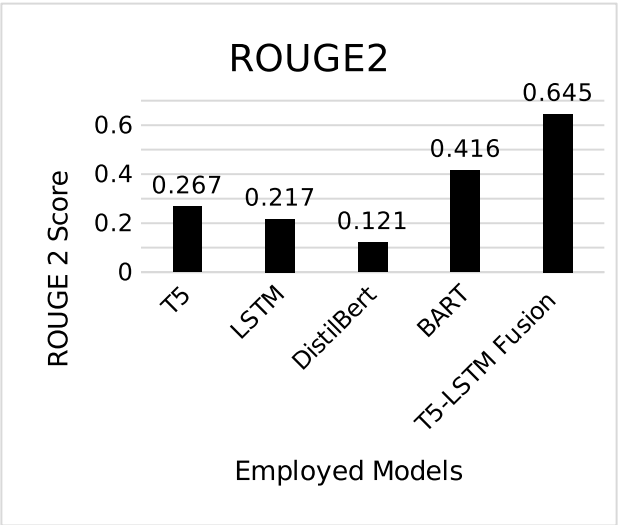

FIGURE 4. ROUGE-2 Score Comparison Across the Proposed and Other Employed Models.

relevant summaries. However, the T5-LSTM FusionNet version outperformed all others with a better score of 0.862, indicating its good performance in producing accurate and relevant summaries.

Figure 4 presents the ROUGE-2 scores, which measure bigram overlap and as a result, provide insight into the fluency and coherence of the summaries, displaying a distinctive score. T5 scored 0.267, improving its relative performance in comparison to ROUGE-1. LSTM's performance, with a score of 0.217, becomes lower than T5's. DistilBert's score dropped extensively to 0.121, suggesting demanding situations in keeping phrase pairs. BART excelled with a score of 0.416, reflecting its strong text generation talents. The T5-LSTM FusionNet model endured to steer with a score of 0.645, emphasizing its better fluency and coherence in summary generation.

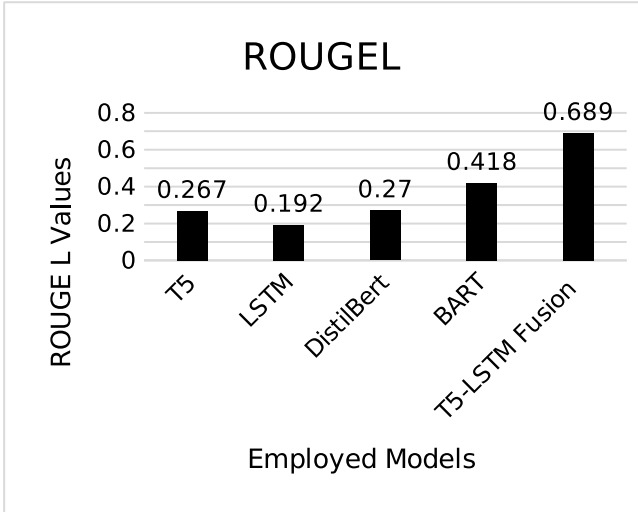

FIGURE 5. ROUGE-L score comparison across the proposed and other employed models.

Figure 5 shows the ROUGE-L score where T5 scored 0.267, and LSTM covered in the back at 0.192. DistilBert's score was barely better at 0.27. BART verified good performance again with a score of 0.418. The T5-LSTM FusionNet version got a better score of 0.689, showing its incredible potential to generate structurally similar and contextually appropriate summaries.

The T5-LSTM FusionNet version continually outperformed the alternative fashions throughout all three ROUGE metrics, indicating its better capability to generate correct, fluent, and coherent summaries. BART additionally completed nicely, mainly in ROUGE-2 and ROUGE-L, highlighting its strength in generating contextually rich and structurally sound summaries. DistilBert, at the same time as effective in unigram overlap, struggled with bigram coherence and structural similarity. The standalone T5 and LSTM models confirmed mild performance, with T5 commonly outperforming LSTM. These results endorse that combining the strengths of various models, as visible within the T5-LSTM FusionNet approach, can lead to extensive enhancements in summarization quality.

Figure 6 illustrates the precision analysis which shows how some of the generated summaries are relevant. T5 and BART have precision rankings of 0.49, indicating that approximately half of their summaries are applicable. DistilBert performs barely better with a precision rating of 0.53. LSTM shows a large development with a precision score of 0.67, suggesting it generates extra applicable summaries. The T5-LSTM FusionNet achieves a better precision score of 0.72, reflecting its capacity to generate relatively relevant summaries. Recall measures the ability of the models to seize all relevant records from the reference summaries as shown in Figure 7. The T5 has a recall score of 0.49, indicating it captures much less than half of the relevant information. DistilBert plays barely well with a considered score of 0.52, and BART follows closely with a rating of 0.51. LSTM shows

TABLE 4. Sample's summarization results using T5-LSTM fusionnet model on psychological dataset.

Original Text	Summaries
The patient has been experiencing prolonged periods of anxiety and mood fluctuations, particularly manifesting in social settings. They report difficulty sleeping and a pervasive sense of hopelessness. The patient's recent experiences with social withdrawal have exacerbated their feelings of isolation, and they often find it challenging to engage in daily activities. During therapy sessions, the patient expressed concerns about their self-worth and an overwhelming fear of failure. This persistent state of distress has impacted their work and relationships, leading to further emotional challenges.	The patient is struggling with anxiety, mood fluctuations, and social withdrawal, resulting in sleep issues and feelings of hopelessness. Concerns about self-worth and fear of failure are affecting their work and relationships, leading to significant distress.
The client describes a recurring pattern of negative thoughts, often centered on self-criticism and fear of judgment from others. This has led to avoidance of social interactions and reluctance to pursue personal or professional goals. They frequently experience racing thoughts and have found it increasingly difficult to concentrate on tasks. The patient's family has expressed concern over these changes, noticing a marked decrease in the patient's engagement with their usual activities and interests.	The client faces persistent negative thoughts, self-criticism, and social avoidance, impacting their motivation and focus. Family members have observed decreased engagement in the client's usual interests.

a better development with a recall score of 0.66, indicating its effectiveness in achieving good performance.

The T5-LSTM FusionNet excels once more with a recall rate of 0.72, demonstrating its better capability in capturing the maximum of the relevant content. The F1-rating, which is the harmonic suggestion of precision and recall, provides a balanced measure of each model's overall performance as presented in Figure 8. T5 has an F1-score of 0.49, indicating balanced however modest overall performance. DistilBert has an F1-score of 0.52, showing slight batter. BART ratings are 0.5, indicating balanced but restricted performance. LSTM achieves a better F1-score of 0.66, indicating its effectiveness in balancing precision and recall. The T5-LSTM FusionNet model, with an F1 score of 0.71, demonstrates the quality of overall better performance, effectively balancing precision and recall.

Accuracy measures the general correctness of the summaries produced through each employed model. Figure 9 presents the accuracy score of each model compared with the proposed model. The T5 and BART each show an accuracy of 0.5, indicating that they efficaciously summarize the text 1/2 of the time. DistilBert plays slightly well with an accuracy of 0.53, reflecting a modest improvement. LSTM achieves a good accuracy of 0.67, displaying its effectiveness in producing accurate summaries. The T5-LSTM FusionNet model outperforms other models with a better accuracy of 0.74, demonstrating its advanced functionality in generating correct summaries. Table 5 presents the percentage difference in precision, recall, F1-score, and accuracy of each model when compared to the proposed model. The analysis indicates that the proposed model and LSTM exhibit only slight variations, with percentage differences of 7.19% in precision, 8.70% in recall, 7.30% in F1-score, and 9.93% in accuracy. These small differences suggest that LSTM is closely aligned with the performance of the proposed model, highlighting its

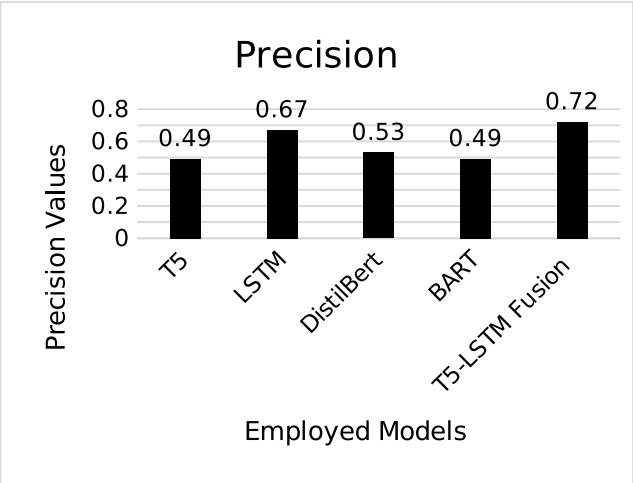

FIGURE 6. Precision analysis of each employed model contrasting with the proposed model.

effectiveness in text summarization tasks. On the other hand, the T5, BART, and DistilBert models show significantly larger percentage differences across these metrics, indicating a more pronounced gap in their performance relative to the proposed model. This disparity underscores the better of the proposed model in terms of its precision, recall, F1-score, and accuracy, positioning it as a more robust and reliable solution for text summarization.

The cross-comparison as shown in Table 6 reveals a clear hierarchy in model performance across ROUGE scores. The values represent the ratio of average ROUGE scores of the model in the row compared to the model in the column, with a value greater than 1 indicating better performance. The T5-LSTM FusionNet model consistently outperforms all other models, as all its values are greater than 1,

FIGURE 7. Recall analysis of each employed model contrasting with the proposed model.

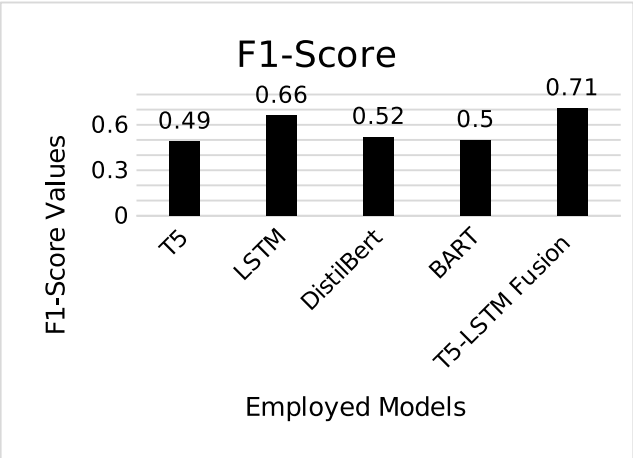

FIGURE 8. F1-Score analysis of each employed model contrasting with the proposed model.

showcasing its effectiveness in combining the strengths of T5 and LSTM architectures. BART ranks as the second-best model, significantly outperforming T5 and LSTM, but it is still outperformed by the T5-LSTM FusionNet, indicating that while BART is robust, it doesn't match the ensemble model's effectiveness. DistilBert, though stronger than LSTM, falls short when compared to T5 and BART, particularly in ROUGE-2, indicating limitations in handling more complex summarization tasks. T5 outperforms LSTM across all metrics but remains less effective than BART and the T5-LSTM FusionNet model.

Table 7 presents the cross-comparison of model accuracies which provides a comprehensive insight into the relative performance of each model. The diagonal values of 1.00 represent each model compared to itself, while values greater than 1 indicate that the model in the row outperforms the model in the column, and values less than 1 indicate the opposite. For example, LSTM's accuracy is 33% higher than

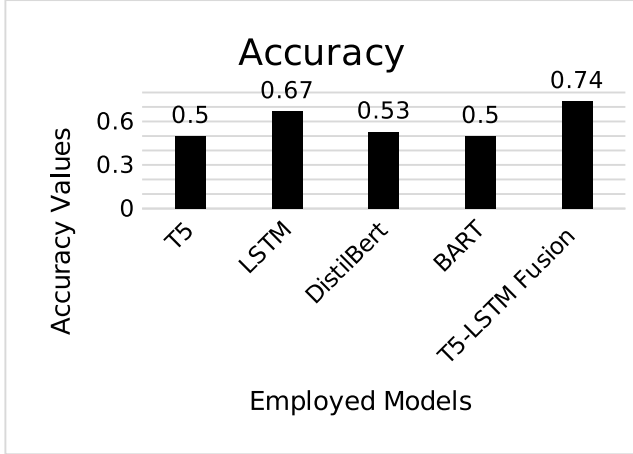

FIGURE 9. Accuracy analysis of each employed model contrasting with the proposed model.

T5's, as shown by the 1.33 ratio in the LSTM row and T5 column. Conversely, T5's accuracy is 32% lower than the T5-LSTM FusionNet model, as indicated by the 0.68 ratio within the T5 row and T5-LSTM FusionNet column. The T5-LSTM FusionNet model emerges because of the maximum robustness, continuously outperforming all different models with values greater than 1 across its row, demonstrating the effectiveness of the ensemble method in leveraging the strengths of both T5 and LSTM architectures. LSTM also indicates sturdy overall performance, considerably outperforming T5, BART, and DistilBERT, with ratios of 0.33, 1.33, and 1.26, respectively. In contrast, BART and DistilBERT, even as more potent than T5, fall quickly when compared to LSTM and T5-LSTM FusionNet, underscoring their limitations in attaining excessive accuracy. T5, though competitive, lags in the back of the opposite fashions, especially the T5-LSTM FusionNet and LSTM, as evidenced with the aid of its decreased ratios. Overall, the table virtually illustrates the prevalence of the T5-LSTM FusionNet model in phrases of accuracy, followed with the aid of LSTM, whilst highlighting the restrictions of T5, BART, and DistilBERT in comparison.

Concerning several criteria including Precision, Recall, F1-Score, Accuracy, and ROUGE scores, the results analysis presented provides a thorough assessment of multiple models used for ATS. The T5-LSTM FusionNet model was introduced to fuse the best aspects of sequential and transformer-based models. Its performance is evaluated rigorously in comparison to the most advanced models. With all scores at 0.49, the T5 model performs consistently but poorly according to the Precision, Recall, and F1-Score criteria. Due to T5's dependency on transformer design and lack of sufficient fine-tuning for summarizing tasks, this consistency suggests consistent but average summarization quality. With scores of about 0.66, the LSTM model performs much better than T5, thanks to its ability to handle sequential data efficiently. Its unexpectedly poor performance in bigram

TABLE 5. T5-LSTM FusionNet percentage difference comparison with employed models.

Models	Percentage Difference in			
	Precision	Recall	F1 Score	Accuracy
T5-LSTM FusionNet with T5	38.02%	38.02%	36.67%	38.71%
T5-LSTM FusionNet with LSTM	7.19%	8.70%	7.30%	9.93%
T5-LSTM FusionNet with DistilBert	30.40%	32.26%	30.89%	33.07%
T5-LSTM FusionNet with BART	38.02%	34.15%	34.71%	38.71%

TABLE 6. Cross-comparison of average ROUGE scores showing the relative performance of models, where the T5-LSTM FusionNet consistently outperforms all others, followed by BART, with DistilBert, T5, and LSTM ranking lower in effectiveness.

Model Comparison	T5	LSTM	DistilBert	BART	T5-LSTM FusionNet
T5	-	0.87	0.60	0.46	0.32
LSTM	1.15	-	0.69	0.53	0.37
DistilBert	1.67	1.44	-	0.84	0.53
BART	2.17	1.89	1.19	-	0.69
T5-LSTM FusionNet	3.15	2.72	1.88	1.45	-

TABLE 7. Cross-comparison of model accuracies, where the T5-LSTM FusionNet model consistently outperforms all others, followed by LSTM, while BART, DistilBERT, and T5.

Model	T5	LSTM	DistilBERT	BART	T5-LSTM FusionNet
T5	1.00	0.75	0.94	1.00	0.68
LSTM	1.33	1.00	1.26	1.33	0.91
DistilBERT	1.07	0.79	1.00	1.07	0.72
BART	1.00	0.75	0.94	1.00	0.68
T5-LSTM FusionNet	1.47	1.10	1.38	1.47	1.00

capture (ROUGE-2) indicates that it may not be able to capture more intricate connections in the text.

The DistilBERT and BERT models outperform the T5 model in terms of accuracy and recall, with the former obtaining somewhat higher scores. They still fall short of the required standard for excellent summary, though. BERT demonstrates very high ROUGE scores (0.856, 0.816, and 0.818) with its considerable pre-training, demonstrating exceptional competence in sequence organization and content retention. The accuracy of 0.5, however, indicates that overfitting or a failure to generalize in many settings may occur even with BERT's excellent recall and precision.

With the greatest precision, recall, and F1 scores (0.72, 0.72, and 0.71, respectively), the T5-LSTM FusionNet model shows how summarization accuracy may be improved by combining transformer-based contextual awareness with LSTM's sequential processing. Its performance in the ROUGE-L (0.189) and ROUGE-2 (0.045) metrics, however, indicates that although it can capture unigrams with ease, it has trouble handling bigrams and preserving sequence integrity. This disparity suggests that to fully utilize the advantages of both component models, the fusion design has to be further refined. The fusion model is superior to the T5 and LSTM models individually, as seen by accuracy analysis, which gives it a maximum accuracy of 0.74. This accuracy shows that, while not totally, the hybrid method tackles some of the shortcomings of standalone models. The performance of the fusion model shows that the combination manner can be reinforced or that further tweaking may be essential to boom bigram and sequence retention. Even while the T5-LSTM Fusion version makes use of sequential models and transformers to its advantage and produces encouraging consequences, there is constantly room for development, especially in terms of handling difficult textual content interactions and guaranteeing sturdy sequence retention. To truly reflect actual-global overall performance and save you overfitting or misleading measures, extra stringent evaluation procedures are required, as proven by way of the found inconsistencies and best ratings in certain fashions.

The higher performance of the T5-LSTM FusionNet model in textual content summarization is broadly speaking because of its hybrid method, which capitalizes on the strengths of both transformer-based and sequential models. By integrating the T5 model's better language understanding and generative capabilities with the LSTM model's scalability in capturing sequential dependencies, the FusionNet achieves a greater nuanced and comprehensive summarization. The T5 issue excels in contextual understanding, producing coherent and

contextually applicable summaries, whilst the LSTM issue correctly manages sequential facts throughout textual content segments. This combination allows FusionNet to generate summaries that aren't only fluent and accurate but also hold the sequential integrity of the unique content material. The fusion of T5's context vector with LSTM's hidden state vector whether or not via concatenation, averaging, or a neural network-based method enables a deeper integration of contextual and sequential information, ensuing in summaries that replicate both the means and shape of the source textual content.

The T5-LSTM FusionNet model has strong potential to summarize complex psychological data, although it is highly computationally complex. The combination of transformer layers in T5 with LSTM sequential processing results in very high memory and processing demands. It may also lead to overfitting, especially when smaller domain-specific datasets are used. Addressing those challenges, optimization techniques and regularization methods require a great exploration for ensuring efficiency and reliability across diverse datasets. The fusion mechanism in T5-LSTM FusionNet preserves sequence integrity by leveraging LSTM to capture sequential dependencies, while T5's attention mechanism ensures contextual understanding. This integration enables LSTM to handle sequential data effectively, without sacrificing the rich, context-aware representations provided by T5, thus maintaining both sequence order and contextual coherence in the summaries.

V. CONCLUSION

Text summarization is critical in managing the ever-increasing volume of textual material available online, allowing for more effective information retrieval and understanding. This observation investigated numerous tactics in ATS, with a unique consciousness of psychology, in which summarizing complex textual material is essential for medical decision-making and studies synthesis. The study proposed a new hybrid model, T5-LSTM FusionNet, that combines the advantages of transformer-based language modeling T5 with LSTM's sequential language capabilities. This model was tested against the contemporary methods inclusive of T5, LSTM, BERT, and DistilBERT, the usage of stringent evaluation criteria such as precision, recall, F1-score, accuracy, and ROUGE. The consequences show that, while current models which include T5 and BERT carry out properly in some regions, they frequently fall quickly in others, specifically when dealing with complicated psychological content. LSTM mentioned for its capacity to understand sequential dependencies, was completed better however tried with better semantic comprehension. DistilBERT achieved competitively however lacked depth in text understanding when compared to its bigger sibling, BERT. In assessment, the T5-LSTM FusionNet model proved to be a promising solution, with better precision, recall, F1-score, and accuracy metrics than an individual model. It efficiently integrates transformers' contextual cognizance with the sequential processing competencies of LSTMs, removing the various drawbacks determined in the standalone model. However, there are still issues in optimizing the fusion process to improve bigram and sequence retention, resulting in better overall summarization quality. This study helps to advance ATS techniques, notably in the field of psychology, by presenting a personalized strategy that improves summarization accuracy and efficacy.

Future research can enhance summarization models by leveraging hybrid approaches like T5-LSTM FusionNet to effectively address the diverse needs of psychological data analysis and clinical decision support systems. There is a pressing need to develop specialized language models tailored specifically for the psychological domain, explore multilingual applications, and refine assessment metrics to better evaluate summarization quality. Additionally, future work will explore hybrid architectures, such as combining T5 with GRU or RoBERTa, to reduce computational demands while enhancing performance. Expanding the model to diverse psychological and behavioral datasets will assess its adaptability across mental health contexts. Models like GPT and RoBERTa were excluded due to their focus on text generation, but advanced ensemble models will be investigated to improve results.

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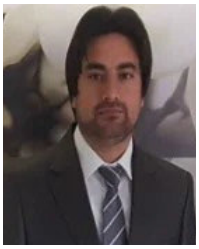

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